

WAQAS UR REHMAN

Software Engineer | Full-Stack & Desktop Applications | React · TypeScript · Node.js · Electron

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PROFESSIONAL PROFILE

Software Engineer with 5+ years building and shipping production software for web and Windows desktop, specializing in React, TypeScript, Node.js and Electron. Currently delivering commercial retail-tech and education-tech products as an independent engineer — three systems live and in daily client use. Strong in offline-first architecture (SQLite, IndexedDB), hardware integration (ESC/POS thermal printers, barcode and QR scanners), REST API development and high-performance UI. Own the full lifecycle: requirements, UI/UX, build, security and data-integrity audits, deployment and client training.

KEY PRODUCTS

Arsh Garments POS & Inventory Suite — Retail POS & Inventory Software (Windows Desktop)

Full-Stack & Desktop Application Engineer | Live in production at client retail outlet

React.js · TypeScript · Node.js · SQLite · Electron · Tailwind CSS · ESC/POS

- Engineered a real-time POS counter in React and TypeScript handling multi-item billing, discounts and tender calculation, with live SKU-level stock deduction on every sale.
- Architected an offline-first SQLite data layer via a Node.js service tier, keeping billing and inventory fully operational during internet outages.
- Integrated thermal receipt printers over the raw ESC/POS protocol, delivering one-click aligned printing without third-party print middleware.
- Built a Customer Credit (Udhar) Ledger and a live Zakat calculation engine, digitising two manual accounting workflows into audited, real-time modules.
- Implemented WhatsApp/SMS gateway integration via REST APIs and a JSON schema-driven backup and restore module with import validation.

Al Hafiz Zaida Barcode Studio — Barcode & Label Generation Software (Desktop & Web)

Frontend & Desktop Application Engineer | Live in production at client wholesale business

React 19 · TypeScript · HTML5 Canvas · Electron · Electron-Builder · JsBarcode · jsPDF

- Packaged a React and Vite application into an offline-first Windows executable using Electron-Builder, shipping both portable and NSIS installer builds for non-technical users.
- Designed a 100% client-side LocalStorage architecture, removing all server, database and internet dependencies and eliminating recurring infrastructure cost.
- Engineered a hardware-aligned print engine with a custom CSS @page stylesheet, overriding default A4 sizing for user-defined label dimensions and resolving landscape-to-thermal-roll rotation conflicts.
- Developed a real-time HTML5 Canvas label preview with dynamic EAN-13 and Code 128 generation, plus a recursive auto-fit algorithm that prevents text overflow on long product names.
- Automated release builds with a custom Node.js script using sandboxed temp directories, resolving Windows file-locking (EPERM) failures that were blocking deployment.

ALS Library Management System — Local-First Library Automation Platform

Lead Frontend Developer | American Lycetuff DNK STEM School (Swabi) | Live in daily campus operations

React 19 · TypeScript · Dexie.js (IndexedDB) · Zustand · jsQR · jsPDF · Recharts · Zod

- Engineered a local-first architecture on Dexie.js and IndexedDB enabling full offline operation with instant state synchronisation and zero network latency.
- Integrated web-camera QR scanning with jsQR to automate issuance, renewal and returns, cutting average transaction time by over 80%.
- Built a bulk-import pipeline with PapaParse and Zod schema validation plus duplicate detection, ensuring 100% data integrity during legacy catalogue migration.
- Implemented multi-role RBAC portals (librarian, focal person, teacher, student) with client-side PDF label generation and Recharts analytics dashboards.
- Created an encrypted backup engine with custom binary .lms exports, securing student and catalogue records against data loss.

PROFESSIONAL EXPERIENCE

Freelance Software Engineer — Independent Consultant

06/2024 – Present | Swabi, Pakistan (remote & on-site clients)

- Deliver custom desktop and web products end to end for retail, wholesale and education clients — requirements, architecture, development, deployment and support.
- Integrate business hardware (ESC/POS printers, label printers, camera-based scanners) and run data-integrity and security audits on live deployments.
- Train non-technical end users and maintain versioned releases with ongoing feature enhancement.

Computer Science Teacher — American Lycetuff DNK STEM School

08/2024 – Present | Swabi, Pakistan

- Teach programming fundamentals, web technologies and computer literacy to secondary-level STEM students through project-based instruction.
- Identified operational gaps in campus library workflows and delivered the ALS Library Management System as the in-house solution now used daily.

Web Developer — WQ Softwares

08/2021 – 06/2024 | Peshawar, Pakistan

- Built responsive, cross-browser user interfaces in React.js and translated design mockups into production-ready components.
- Developed and deployed RESTful APIs with Node.js and Express.js for reliable client-server data flow.
- Designed and optimised MongoDB schemas and queries, improving application performance across client projects.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3, SQL, C++

Frontend: React.js / React 19, Vite, Tailwind CSS, Bootstrap, HTML5 Canvas API, Responsive Design, UI/UX, Component Architecture

Backend & APIs: Node.js, Express.js, REST APIs, JSON Schema Design, Third-Party Gateway Integration (WhatsApp, SMS)

Data & State: SQLite, MongoDB, IndexedDB (Dexie.js), LocalStorage, Zustand, React Hooks, State Management, Query Optimisation

Desktop & Hardware: Electron, Electron-Builder, NSIS Packaging, Offline-First Architecture, ESC/POS Thermal Printing, Barcode & QR Scanning

Practices: Security Audits, Data Integrity Validation, Backup & Recovery, Performance Optimisation, Git & GitHub, Agile Delivery, Technical Documentation

EDUCATION

BS Software Engineering — Pak-Austria Fachhochschule, Institute of Applied Sciences and Technology (PAF-IAST), Haripur, Pakistan | 2020 – 2024

CERTIFICATIONS & TRAINING

- **IBM Full Stack Software Developer Professional Certificate** — IBM
- **Full Stack Developer Bootcamp: Master Frontend to Backend** — GeeksforGeeks
- **Web MERN Stack Development** — Chai aur Code
- **Introduction to JavaScript** — Great Learning
- **Introduction to NoSQL** — Great Learning
- **Mastering User Interface and Design** — Coursera
- **AI for Everyone** — Coursera
- **Foundations of Project Management** — Google
- **User Interface and User Experience (UI/UX) Training** — PAF-IAST
- **Frontend Web Development Workshop** — TechRex

LANGUAGES

English: Professional working proficiency (CEFR B2) | **Arabic:** Basic